FoodArrival

Workflow and architecture overview

Product summary

Multi-role food ordering and delivery application.
ASP.NET MVC 5 on .NET Framework 4.8.1, with Razor UI, C#,
ADO.NET repositories, and SQL Server stored procedures.

Main capabilities

Customer browsing, cart, order placement, payment simulation, tracking
Restaurant and kitchen order preparation and menu operations
Rider assignment, pickup, delivery progress, and OTP handoff
Admin dashboards, approvals, transactions, and bulletins

System boundary

Payment is simulated. ETA, map coordinates, and route logic include
demo/fallback data and should not be treated as live integrations.

Application architecture

Request path and main component responsibilities

Layer responsibilities

Browser UI: Razor views, Bootstrap, jQuery and AJAX

MVC: route dispatch, portal controllers, validation, session role checks

Services: rider allocation and order tracking calculations

DAL: repositories plus DbHelper for SQL calls and result mapping

Database: FoodArrivalDB tables, stored procedures, and order history

Main project areas

Controllers/ | request coordination and role workflows

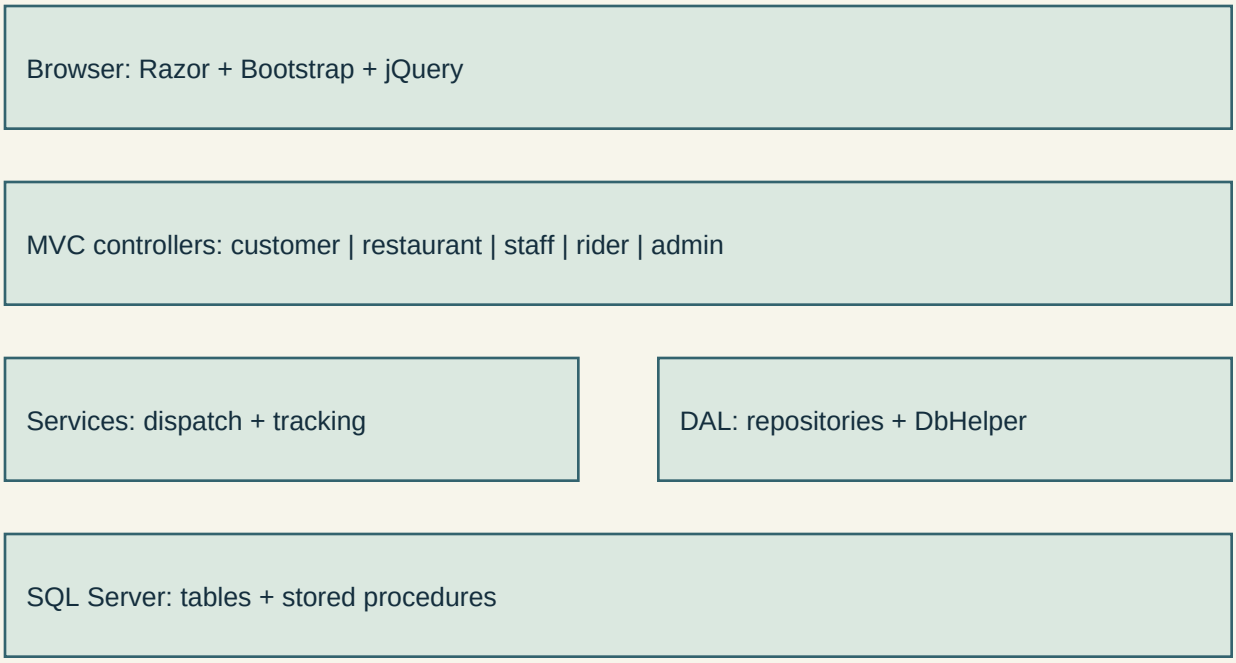
Services/ | allocation and tracking logic

DAL/ | database connectivity and repositories

Models/ | domain types and view models

Views/ | customer, restaurant, staff, rider, admin pages

App_Start/ | route, bundle, and global filter registration



Order and delivery workflow

Master order may be split into restaurant-specific sub-orders

Customer and fulfillment journey

1. Customer browses menu, builds cart, and submits delivery details.
2. Application creates an order and its line items/sub-orders.
3. Payment simulation records payment/transaction outcome.
4. Restaurant accepts and prepares its assigned sub-order.
5. Ready order is considered for rider assignment and pickup.
6. Rider updates pickup and delivery progress.
7. Customer provides the delivery OTP to confirm handoff.
8. Order history and tracking show current state and status events.

Other role loops

Restaurant: maintain menu process kitchen queue mark ready.

Rider: review assignment pick up update status confirm delivery.

Admin: review operations approvals and transactions publish bulletins.

Notes for operators

Set the FoodDeliveryDB connection string in Web.config. Initialize FoodArrivalDB with the consolidated script or matching SQL scripts.

Use secure secrets; replace embedded local fallback credentials.